The Island of Truth

Immanuel Kant, *Critique of Pure Reason* (1781)

A land of truth in a sea of illusion

C. D. Friedrich,
Wanderer above the Sea of Fog (c. 1817)

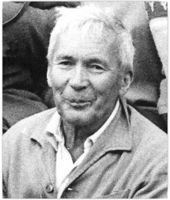
Es ist das Land der Wahrheit (ein reizender Name), umgeben von einem weiten und stürmischen Oceane, dem eigentlichen Sitze des Scheins, wo manche Nebelbank und manches bald wegschmelzende Eis neue Länder lügt …

It is the *land of truth* (an enchanting name), surrounded by a wide and stormy ocean, the very *seat of illusion*, where many a *fog bank* and many a swiftly melting iceberg *feign new lands* …

Kritik der reinen Vernunft, A235 / B294 (1781 / 1787)

To seek the truth is to know the fog for fog, and never chart land that is not there.

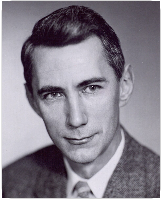
Two ways to measure the fog



Andrey Kolmogorov (1903 to 1987)

Probability theory

variance: how far outcomes spread around the average



Claude Shannon (1916 to 2001)

Information theory

entropy: the average surprise, in yes/no questions

Entropy and variance: two closely linked measures of the same fog.

The formulas behind the weekend

Information theory (Shannon)

$$H(X) = -\sum_x p(x)\,\log_2 p(x)$$

entropy: the average surprise, in bits

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

mutual information: what X tells you about Y

$$D_{\mathrm{KL}}(p\,\|\,q) = \sum_x p(x)\,\log_2 \frac{p(x)}{q(x)}$$

relative entropy: the cost of believing q, not p

Probability theory (Kolmogorov)

$$\mathrm{Var}\Big(\sum_i X_i\Big) = \sum_i \mathrm{Var}(X_i) + 2\sum_{i<j}\mathrm{Cov}(X_i, X_j)$$

spread of a total: the spreads, plus how the parts move together

$$\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n} + \Big(1 - \tfrac{1}{n}\Big)\bar{\rho}\,\sigma^2$$

averaging shrinks the spread, unless the parts agree

Roadmap of the week

1. **The Gaussian distribution**
   the bell curve and its spread
2. **Information theory**
   entropy, the bit, mutual information
3. **Information-theoretic active learning**
   ask the most informative question
4. **Variance reduction with ensembles**
   combine many models to cut the spread
5. **Competitive machine learning**
   win the leaderboard: XGBoost and Kaggle